Davide Capone

Date of birth: 23/04/1999 **Nationality**: Italian

CONTACT INFORMATION

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PROFILE DESCRIPTION

I am a Data Science and Scientific Computing student with a growing foundation in artificial intelligence, data analysis, and scientific problem-solving. My academic journey began with a bachelor's degree in *Internet of Things, Big Data & and Web*, where I gained valuable insights into modern technologies and their applications. I am currently pursuing a master's degree in *Data Science and Scientific Computing*, focusing on advanced data analysis, artificial intelligence, and computational methods. I have also had the opportunity to develop my communication and presentation skills as a *Computer Science teacher* at a secondary school, enabling me to explain technical concepts effectively to diverse audiences. I am motivated by a genuine passion for computer science and mathematics, and I am eager to contribute to innovative AI projects, continue learning, and grow as a professional in the field.

EXPERIENCE

Deep Learning Engineer (Intern)
Danieli Research Center, Buttrio (Udine), ITA 2024/12 → Present

What I am doing here?

- Building and optimizing existing deep learning models by leveraging state-of-the-art algorithms to improve performance and accuracy.
- Studying and utilizing machine learning frameworks tailored for computer vision tasks, ensuring efficient implementation and scalability.
- Researching and implementing advanced object tracking algorithms to address specific industrial challenges.
- Collaborating with team members to integrate optimized models into operational systems and evaluate their performance in production environments.

Computer Science Teacher
ISIS Paschini Linussio, ISIS Raimondo D'Aronco, (Udine), ITA 2023/10 → 2024/09

- Taught computer science, programming, and ICT technologies to secondary school students.
- Designed lesson plans combining theory and practice to ensure a comprehensive understanding of core topics.
- Fostered collaboration and critical thinking through group projects, coding challenges, and real-world problem-solving.

Computer Vision Engineer (Intern)
Danieli Research Center, Buttrio (Udine), ITA 2022/10 → 2022/12

- Computer vision techniques, including calibration methods such as homography estimation for camera alignment, object detection and object localization, with a focus on addressing real-world industrial challenges.
- Researched and implemented state-of-the-art algorithms to optimize detection and localization of objects in a steel slab storage yard.
- Created datasets using motion detection algorithms and YOLO for bounding box management, tailored to industrial-scale requirements.
- Designed, implemented, and tested convolutional neural networks (CNNs) for object localization and detection.

- The analysis of model performances with hyperparameter tuning methods was undertaken, with the objective of identifying and deploying the most effective solutions in a production setting.

Education

Master's Degree in Data Science and Scientific Computing
University of Trieste, (Trieste), ITA 2023/10 → Present

The course is jointly offered by the *University of Trieste*, *University of Udine*, *ICTP* and *SISSA*.

- Key courses: *Advanced Programming and Algorithmic Design, Data Management for Big Data, High Performance Computing, Numerical Analysis, Probabilistic Machine Learning, Reinforcement Learning, Statistical Methods for Data Science, Deep Learning, Information Retrieval and Data Visualization.*

Bachelor's Degree in Internet of Things, Big Data & Web
University of Udine, (Udine), ITA 2019/10 → 2022/12

Final Grade: 100/110 — Thesis: *Applications of Computer Vision for Object Detection and Localization in a Steel Yard Plant.*

- Key courses: *Algorithms and Data Structures, Machine Learning, Data Science, Applied Statistics, Social Computing, Mathematical Analysis, Social Computing, Object Oriented Programming, Cloud Computing, Software Engineering, Databases.*

Projects

GAPC - Asteroid Observation Catalog. GAPC is a Django-based application designed to manage and catalog asteroid observations, including metadata from associated FITS files. It implements astronomical standards, such as VOTable, to facilitate interoperability with other astronomical datasets and tools. Originally developed for the Gruppo Astronomico della Polse di Cougnes, GAPC is tailored to meet the needs of its members in managing asteroid observation data. The platform is designed to be extensible, making it suitable for any group or organization conducting asteroid research and observation. Users can explore a catalog of asteroids, retrieve observation data, and download metadata in VOTable format, supporting linked open data principles.

YOHO24. YOHO24 is a neural network model for audio segmentation and sound event detection tasks, inspired by the original YOHO model by Venkatesh S. et al. This project was part of the final exam for the Deep Learning course at the University of Trieste during the 2023-2024 academic year. The project aimed to improve the performance of the YOHO algorithm by applying features introduced in more recent YOLO models.

Weighted Matrix Factorization Model. Developed a recommender system using the MovieLens 100k dataset. Implemented matrix factorization techniques optimized with Stochastic Gradient Descent (SGD) and Weighted Alternating Least Squares (WALS). The project includes detailed analysis and comparison of these optimization algorithms.

Traffic Lights Optimization. Simulated an intersection with traffic lights and vehicles using Pygame. Applied Reinforcement Learning techniques, including Policy Iteration and Value Iteration, to optimize traffic light policies, aiming to reduce waiting times and queue lengths. The project serves as a final exam for the Reinforcement Learning course at the University of Trieste.

Selection Algorithms Analysis. Analyzed and implemented selection algorithms, focusing on variations of the 'quick sort' algorithm. The project includes a C program that demonstrates these algorithms and provides performance comparisons.

Languages

Italian: Mother tongue.

English: Listening (B2), Reading (B2), Writing (B2).

Driving License

Driving Licence: B.